DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer – 2023

Course: B. Tech. **Branch : Electronics and Communication Engineering / Electronics and Telecommunication Engineering**

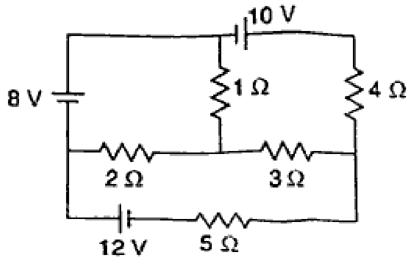
Subject Code & Name: Network Theory (BTETC401) **Semester : IV**

Max Marks: 60 **Date:13/07/2023** **Duration: 3 Hr.**

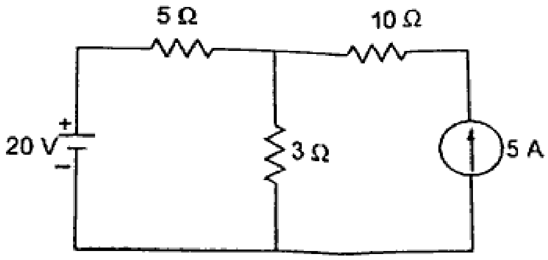
Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

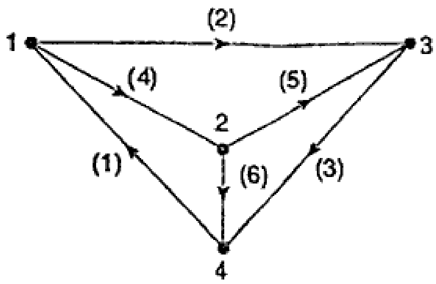
		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	State Kirchhoff's laws and Determine the current through the 5 Ω resistor of the network shown	**CO1**	**6**



B)	State superposition theorem and find the current passing through the 3 Ω resistor in the circuit using superposition theorem.	**CO1**	**6**

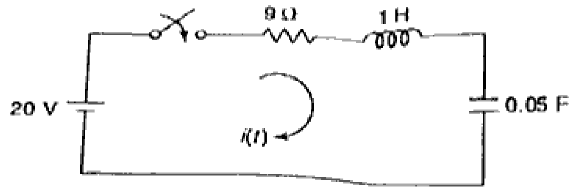


C)	The graph of a network is shown in Fig. Write the (a) incidence matrix, (b) tieset matrix, and (c) f-cutset matrix.	**CO1**	**6**



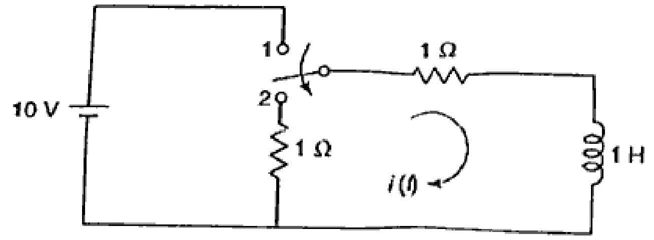
Q.2	**Solve Any Two of the following.**		**12**
A)	What are initial conditions? Explain the initial conditions for Resistor, capacitor and inductor	**CO1, CO3**	**6**
B)	In the given network of Fig., the switch is closed at t = 0. With zero current in the inductor, find i, di/dt and $d^2 i/ dt^2$ at t = 0+ .	**CO1, CO3**	**6**

C) In the network of Fig. shown, the switch is closed at t 0+. Obtain the expression for current i (t) (use time domain approach) CO1, CO3 6



Q. 3 Solve Any Two of the following. 12

A) In the network of Fig. 7.12, the switch is moved from the position 1 to 2 at t = 0, steady-state condition having been established in the position 1. Determine i (t) for t > 0 (Using S-domain) CO1 6



B) To find Initial and Final values if they exist of the signal with L.T. CO1 6

(a) $F(s) = \dfrac{7s^2 + 63s + 134}{(s+3)(s+4)(s+5)}$ (b) $F(s) = \dfrac{4s^2 + 7s + 1}{s(s+1)^2}$

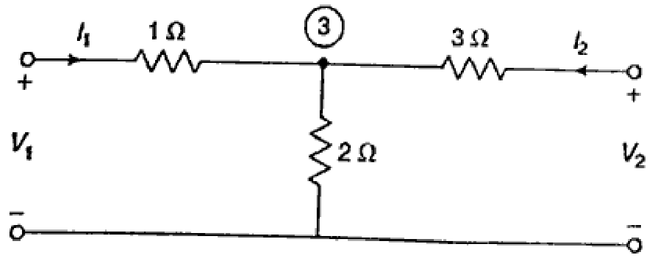
(c) $F(s) = \dfrac{40}{(s^2 + 4s + 5)^2}$

C) Find Laplace Transform of various standard time domain functions CO1 6

Q.4 Solve Any Two of the following. 12

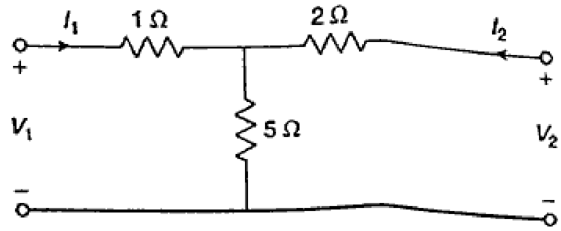
A) Define the terms: Driving-point Impedance Function, Driving-point Admittance Function CO1, CO4 6

Find Y-parameters for the network shown in Fig



B) Derive h -parameter in terms of Z parameter, Y-parameter and ABCD parameter CO1, CO4 6

C) Find the transmission or general circuit parameters for the circuit shown in Fig. Comment on reciprocity and symmetry | CO1, CO4 | 6



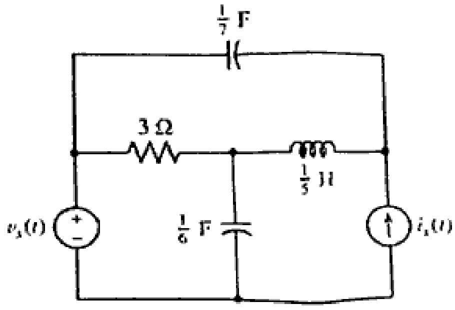
Q. 5 Solve Any Two of the following. **12**

A) Explain about classification of filters. Draw the characteristics of Low-pass and High-pass filters. **CO1, CO3** **6**

B) Realise the following function in Foster-I and Foster-II forms **CO1, CO3** **6**

$$Z(s) = \frac{3(s+2)(s+4)}{s(s+3)}$$

C) Obtain the normal-form equations for the circuit of Fig. shown, a four-node circuit containing two capacitors, one inductor, and two independent sources **CO1, CO3** **6**



*** **End** ***